

Advanced Algorithms

Examination – October 23 2025

Duration: 2 h

Number of page(s): 2.

Only documents from the course or personal notes are permitted. Before you begin, it is recommended to read all the document.

Grading scale (temporary)

Part	1	2	3
on	7 pts	3 pts	10 pts

Part 1 Complexity (7 points)

- Order increasingly the following functions, according to their growth rate $O(\cdot)$:
 $\exp(2)$, $\frac{|\sin(n)|}{n}$, $\frac{n}{2} \log(n)$, $n^{4/3}$, $2^{n/2}$
- Consider the two functions: $f_1(n) = \begin{cases} 2^n, & \text{if } n \leq 3^{10} \\ n, & \text{if } n > 3^{10} \end{cases}$ and $f_2(n) = \begin{cases} n, & \text{if } n \leq 100 \\ n^3, & \text{if } n > 100. \end{cases}$
 Give the true assertion(s) among the following ones: (a) f_1 is $O(f_2)$, (b) f_2 is $O(f_1)$, (c) f_1 and f_2 are incomparable.
- True or false? Justify your answers.
 - If $g(n) \in O(f_1(n))$, $f_1(n) \in O(f_2(n))$ and $h(n) \in O(f_2(n))$ then $g(n) + f_2(n) \in \Omega(h(n))$.
 - If $g(n) \in O(f_1(n))$, $f_1(n) \in O(f_2(n))$ and $h(n) \in O(f_2(n))$ then $g(n) + h(n) \in \Omega(f_1(n))$.
- Solve the following recurrences by giving their asymptotic behavior:
 - $T(n) = 2 * T(n/4) + 2n^3$, for $n \geq 2$ and $T(1) = 1$.
 - $T(n) = 4 * T(n-1) + 1$
 - $T(n) = 2 * T(n/2) + \log_2(n^2)$, for $n \geq 2$ and $T(1) = 1$.

Part 2 Branch and Bound (3 points)

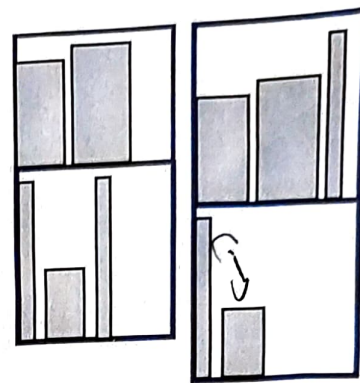
- Recall quickly the principle of Branch and Bound.
- Propose a solution for the Traveling Salesman Problem (TSP) by designing an evaluation function in each node of the search tree such that it can estimate a lower bound on the solution that can be achieved from this node and that is different from the one seen in class. Integrate appropriately this function in the branch and bound algorithm.

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Part 3 The minimum height shelf (10 points)

We would like to store N books $\{B_1, \dots, B_N\}$, in a bookcase with multiple shelves (*i.e.* levels/floors) according to the following constraints:

- The order in which the books are arranged is fixed and must follow the index associated to each book: B_1 must be the leftmost book in the top shelf, B_2 must be located to the right of B_1 , or at the far left of the shelf below, and so on.
- The bookcase, and therefore each of its shelves, has a fixed width L .
- Each book B_i is characterized by its width e_i and its height h_i .



Example with 5 books and 2 bookcases with 2 shelves. On the left books B_1 and B_2 only are placed on the 1st shelf leading to minimal height in comparison to the right where books B_1 to B_3 are placed on the first shelf.

The problem is to figure out how to arrange the shelves so that they have a minimum height.

1. Greedy solution.

- Propose a greedy approach for this problem.
- Can you find a setting where your solution can be optimal? Justify.
- Can you find a setting where your solution is not optimal? Justify.

2. We now denote $\text{hgmin}(i)$ the minimal global height of a shelf of width L on which are stored all the books $B_i, \dots, B_N$ ($1 \leq i \leq N$) and so that $\text{hgmin}(N+1) = 0$ for the height of the shelf where no book is placed. Let us denote $\text{hr}(i, j)$ the height of the shelf where the series of books B_i to B_j is stored. We have that

$$\text{hr}(i, j) = \begin{cases} \max_{k \in \{i, \dots, j\}} \{h_k\} & \text{if } \sum_{k=i}^j e_k \leq L \\ +\infty & \text{otherwise.} \end{cases}$$

Propose a decomposition of $\text{hgmin}(i)$ showing its optimal substructure property.

3. Deduce the general recursive definition of $\text{hgmin}(i)$.

4. Give the pseudo-code of a dynamic programming algorithm able to compute the optimal solution for $\text{hgmin}(i)$.

5. Apply your algorithm on the following data:

i	1	2	3	4	5	6
e_i	1	2	1	1	2	1
h_i	1	2	5	4	3	1

Difficult Now, we propose to fix the width L of the bookcase and the number of shelves K , assuming they have the same width. Given a sequence of N books that do not fit in the bookcase, find an algorithm to find the maximum of books that can fit in the bookcase, knowing that you can remove some books, but you have to keep the initial order.